

CSA0921 - Programming in Java

CO3 - AT2 - ASSIGNMENT 2

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The screenshot shows the SIMATS Online Programming Lab interface. The top navigation bar includes the SIMATS Engineering logo, the lab name, and the user's name 'Vidhushini T S'. The main heading is 'CO3-AT3- ASSIGNMENT 2'. On the left, a 'QUESTIONS' sidebar shows 'Q1 Bank Loan Application Processing System' with a green checkmark and '50 marks'. The main content area for Q1 describes a financial institution's loan application process and lists requirements: a) Create classes for Applicant, LoanApplication, and LoanProcessor; b) Validate applicant name, income, requested loan amount, and credit score; c) Handle built-in exceptions such as NumberFormatException, NullPointerException, and IllegalArgumentException; d) Create user-defined exceptions such as InvalidIncomeException, LoanLimitExceededException, and LoanNotEligibleException; e) Use try-catch-finally during application processing; f) Create multiple threads for processing applications from different branches; g) Use thread priorities for urgent loan applications; h) Use synchronization while updating the common approved-loan database; i) Implement handling for uncaught exceptions and generate a final approval/rejection report. Below the text, the 'Your Code' section shows the beginning of a Java program with imports for java.util.ArrayList and java.util.List, and a Main class with a main method. The 'Input' section contains 'stdin input'. The 'Output' section shows a partial output: '-[Branch: North][ID:APP103] Reason (User Exception): Credit score below minimum requirement of 650'.

This screenshot continues the Java code from the previous one. It includes the creation of two ArrayLists, branch1Apps and branch2Apps, and the addition of LoanApplication objects to them. It then creates two threads, t1 and t2, each running a LoanProcessor for a different branch. The threads are started, and the main thread waits for them to finish using join(). After the threads complete, the main thread prints a report from the ApprovedLoanDatabase. The code also includes user-defined exceptions: InvalidIncomeException and LoanLimitExceededException. The final output in the 'Output' section shows the same message as before: '-[Branch: North][ID:APP103] Reason (User Exception): Credit score below minimum requirement of 650'.

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JAVA PROGRAMMING

```
37 class InvalidIncomeException extends Exception{
38     public InvalidIncomeException(String message){
39         super(message);
40     }
41 }
42 class LoanLimitExceededException extends Exception{
43     public LoanLimitExceededException(String message){
44         super(message);
45     }
46 }
47 class LoanNotEligibleException extends Exception{
48     public LoanNotEligibleException(String message){
49         super(message);
50     }
51 }
52 /*Class Applicant*/
53 class Applicant{
54     private String name;
55     private double income;
56     private int creditScore;
57     public Applicant(String name,double income,int creditScore){
58         this.name=name;
59         this.income=income;
60         this.creditScore=creditScore;
61     }
62     public String getName(){
63         return name;
64     }
65     public double getIncome(){
66         return income;
67     }
68     public int getCreditScore(){
69         return creditScore;
70     }
71 }
72 /*Class LoanApplication*/
73 class LoanApplication{
```

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JAVA PROGRAMMING

```
71 }
72 /*Class LoanApplication*/
73 class LoanApplication{
74     private String applicationId;
75     private Applicant applicant;
76     private double requestedAmount;
77     private int repaymentPeriodMonths;
78     private boolean isUrgent;
79     public LoanApplication(String applicationId,Applicant applicant,double requestedAmount,int repaymentPeriodMonths,boolean isUrgent){
80         this.applicationId=applicationId;
81         this.applicant=applicant;
82         this.requestedAmount=requestedAmount;
83         this.repaymentPeriodMonths=repaymentPeriodMonths;
84         this.isUrgent=isUrgent;
85     }
86     public String getApplicationId(){
87         return applicationId;
88     }
89     public Applicant getApplicant(){
90         return applicant;
91     }
92     public double getRequestedAmount(){
93         return requestedAmount;
94     }
95     public boolean isUrgent(){
96         return isUrgent;
97     }
98 }
99 /*Common approved-loan database with synchronization*/
100 class ApprovedLoanDatabase{
101     private static List<String> approvedLoans=new ArrayList<>();
102     private static List<String> rejectedLoans=new ArrayList<>();
103     public synchronized static void addApproved(String details){
104         approvedLoans.add(details);
105     }
106     public synchronized static void addRejected(String details){
107         rejectedLoans.add(details);
108     }
109 }
```

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JAVA PROGRAMMING

```
104 approvedLoans.add(details);
105 }
106 public synchronized static void addRejected(String details){
107     rejectedLoans.add(details);
108 }
109 public static void printReport(){
110     System.out.println("\n==FINAL APPROVAL/REJECTION REPORT==");
111     System.out.println("APPROVED APPLICATIONS:");
112     if(approvedLoans.isEmpty()){
113         System.out.println("None");
114     }else{
115         for(String app:approvedLoans){
116             System.out.println(" "+app);
117         }
118     }
119     System.out.println("REJECTED APPLICATIONS:");
120     if(rejectedLoans.isEmpty()){
121         System.out.println("None");
122     }else{
123         for(String app:rejectedLoans){
124             System.out.println(" "+app);
125         }
126     }
127 }
128 }
129 /*LoanProcessor class implementation Runnable for multi-threading*/
130 class LoanProcessor implements Runnable{
131     private String branchName;
132     private List<LoanApplication> applications;
133     public LoanProcessor(String branchName, List<LoanApplication> app)
134     {
135         this.branchName=branchName;
136         this.applications=applications;
137     }
138     /*h,e,d,e:Validation,Exception Handling and try-catch-finally*/
139     public void validateAndProcess(LoanApplication app) throws Excepti
140     {
141         try{
142             if(app.getApplicant()==null){
143                 throw new NullPointerException("Applicant data is miss
144             }
145             Applicant applicant=app.getApplicant();
146             if(applicant.getName()==null||applicant.getName().trim().l
147                 throw new IllegalArgumentException("Applicant name can
148             }
149             if(applicant.getIncome()<=0){
150                 throw new InvalidIncomeException("Income must be posit
151             }
152             if(app.getRequestedAmount()<=0){
153                 throw new IllegalArgumentException("Requested loan am
154             }
155             if(app.getRequestedAmount()>applicant.getIncome()*10){
156                 throw new LoanLimitExceededException("Requested loan e
157             }
158             if(applicant.getCreditScore()<650){
159                 throw new LoanNotEligibleException("Credit score below
160             }
161             ApprovedLoanDatabase.addApproved("["+branchName+"]ID:"+app
162             System.out.println("SUCCESS : Application"+app.getApplicati
163         }catch(NullPointerException|IllegalArgumentException e){
164             ApprovedLoanDatabase.addRejected("["+branchName+"]ID:"+app
165             System.out.println("FAILED: Application"+app.getApplicati
166         }catch(InvalidIncomeException|LoanLimitExceededException|Loan
167             ApprovedLoanDatabase.addRejected("["+branchName+"]ID:"+app
168             System.out.println("FAILED: Application"+app.getApplicati
169         }finally{
170             System.out.println("Loan Processing attempt completed for "+
171         }
172     }
173 }
```

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JAVA PROGRAMMING

```
131 private String branchName;
132 private List<LoanApplication> applications;
133 public LoanProcessor(String branchName, List<LoanApplication> app)
134 {
135     this.branchName=branchName;
136     this.applications=applications;
137 }
138 /*h,e,d,e:Validation,Exception Handling and try-catch-finally*/
139 public void validateAndProcess(LoanApplication app) throws Excepti
140 {
141     try{
142         if(app.getApplicant()==null){
143             throw new NullPointerException("Applicant data is miss
144         }
145         Applicant applicant=app.getApplicant();
146         if(applicant.getName()==null||applicant.getName().trim().l
147             throw new IllegalArgumentException("Applicant name can
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149         if(applicant.getIncome()<=0){
150             throw new InvalidIncomeException("Income must be posit
151         }
152         if(app.getRequestedAmount()<=0){
153             throw new IllegalArgumentException("Requested loan am
154         }
155         if(app.getRequestedAmount()>applicant.getIncome()*10){
156             throw new LoanLimitExceededException("Requested loan e
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158         if(applicant.getCreditScore()<650){
159             throw new LoanNotEligibleException("Credit score below
160         }
161         ApprovedLoanDatabase.addApproved("["+branchName+"]ID:"+app
162         System.out.println("SUCCESS : Application"+app.getApplicati
163     }catch(NullPointerException|IllegalArgumentException e){
164         ApprovedLoanDatabase.addRejected("["+branchName+"]ID:"+app
165         System.out.println("FAILED: Application"+app.getApplicati
166     }catch(InvalidIncomeException|LoanLimitExceededException|Loan
167         ApprovedLoanDatabase.addRejected("["+branchName+"]ID:"+app
168         System.out.println("FAILED: Application"+app.getApplicati
169     }finally{
170         System.out.println("Loan Processing attempt completed for "+
171     }
172 }
```

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JAVA PROGRAMMING

```
154         throw new LoanLimitExceededException("Requested loan e
155     }
156     if (applicant.getCreditScore() < 650) {
157         throw new LoanNotEligibleException("Credit score below
158     }
159     ApprovedLoanDatabase.addApproved("["+branchName+"]ID:"+app
160     System.out.println("SUCCESS :Application"+app.getApplicati
161 }catch (NullPointerException | IllegalArgumentException e) {
162     ApprovedLoanDatabase.addRejected("["+branchName+"]ID:"+app
163     System.out.println("FAILED: Application"+app.getApplicati
164 }catch (InvalidIncomeException | LoanLimitExceededException | Loan
165     ApprovedLoanDatabase.addRejected("["+branchName+"]ID:"+app
166     System.out.println("FAILED: Application"+app.getApplicati
167 }finally{
168     System.out.println("Log:Processing attempt completed forAp
169 }
170 }
171 }
172 @Override
173 public void run() {
174     for (LoanApplication app: applications) {
175         try {
176             validateAndProcess(app);
177         } catch (Exception e) {
178             System.out.println("Unexpected exception caught:"+e.ge
179         }
180     }
181 }
```

Run Save

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